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Gravitational Slingshots

Navigating a spacecraft near Gargantua is hard because the speeds are very high. To survive, a planet or star or spacecraft must counteract Gargantua's huge gravity with a comparably huge centrifugal force. This means it must move at very high speed. Near the speed of light, it turns out. In my science interpretation of *Interstellar*, the *Endurance*, parked at ten Gargantua radii while the crew visit Miller's planet, moves at one-third the speed of light: $c/3$, where c represents the speed of light. Miller's planet moves at 55 percent the speed of light, $0.55c$.

To reach Miller's planet from the parking orbit in my interpretation (Figure 7.1), the Ranger must slow its forward motion from $c/3$ to far less than that, so Gargantua's gravity can pull it downward. And when it reaches the vicinity of the planet, the Ranger must turn from downward to forward. And, having picked up far too much speed while falling, it must slow by about $c/4$ to reach the planet's $0.55c$ speed and rendezvous with it.